# Highlights

**GT-Conditioned Variational Gaussian Splatting for Sparse-View Cone-Beam CT Reconstruction**

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- A GT-conditioned CVAE for Gaussian-splatting sparse-view CBCT reconstruction: ground truth is encoded into the variational posterior during training only.
- Deployment discards both the GT encoder and dense views, sampling exclusively from a sparse-view prior; per-point predictive variance comes for free from Monte-Carlo estimation.
- Ablations isolate the decisive design choices: posterior access to the ground truth and fusion token initialization ($\sim 2\,\mathrm{dB}$), with the decoder routed through the context features.

# GT-Conditioned Variational Gaussian Splatting for Sparse-View Cone-Beam CT Reconstruction

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**Abstract**

Sparse-view cone-beam CT (CBCT) reconstruction aims to recover a tomographic volume from an extremely limited set of projection views, which is an ill-posed inverse problem that traditional methods make it difficult to achieve effective reconstruction, and is typically solved with deterministic deep networks. A probabilistic formulation can in principle quantify reconstruction uncertainty and exploit privileged information — dense views or even the ground-truth volume — during training, as long as inference never depends on it. In this work we present the *GT-conditioned CVAE* a conditional variational extension of the 3D-Gaussian-splatting CBCT reconstructor. During training only, a lightweight 3D convolutional network encodes the ground-truth volume into a 32-dimensional descriptor that conditions the variational posterior $q(z \mid C, Y)$ over a 64-dimensional global scene latent, while a sparse-view neural-process prior $p(z \mid C)$ is trained jointly. At deployment the GT encoder and the posterior are discarded: $z$ is drawn exclusively from the sparse-view prior, so inference requires exactly the same inputs as the deterministic backbone and additionally yields per-point predictive variance from $N{=}8$ Monte-Carlo samples. The decoder renders exclusively from the sparse context features, so the latent is the sole channel of dense information during training. On the LUNA16 CBCT benchmark with six projections, the GT-conditioned CVAE reaches 29.50 dB PSNR / 0.868 SSIM, within 0.04 dB of the deterministic backbone (29.54 dB / 0.870 SSIM), while providing calibrated deployment-side uncertainty estimates at no additional inference cost. Ablations quantify the decisive design choices – posterior access to the ground

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truth and fusion token initialization – for future probabilistic CBCT models.

*Keywords:* Sparse-view CBCT reconstruction, 3D Gaussian splatting, conditional variational autoencoder, variational inference, latent neural processes, uncertainty estimation

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## 1. Introduction

Cone-beam computed tomography (CBCT) is a cornerstone of image-guided procedures in radiation therapy and interventional radiology, offering volumetric imaging with compact gantry geometries. Clinical workflows, however, place hard limits on the number of projection views that can be acquired: fewer views reduce radiation dose and shorten scan times, but dramatically undersample the Fourier domain of the object. Sparse-view CBCT reconstruction – recovery of a high-quality volume from a handful of projections, typically far below the hundreds required by analytical reconstruction – is therefore an ill-posed inverse problem whose solution requires strong prior knowledge of anatomy.

Deep learning has become the dominant tool for this task. Neural reconstruction methods range from direct 2D→3D back-projection networks to score-based and diffusion models, and recent work has shown that explicit scene representations such as 3D Gaussian splatting (3DGS) [1] provide an outstanding inductive bias for tomographic volumes: compact, differentiable, and easily supervised by projection consistency. The DIF-Gaussian framework [2] demonstrated that a splatting-based reconstructor can produce accurate volumes from as few as six cone-beam projections.

Despite these advances, the vast majority of sparse-view CBCT models are *deterministic*: they map a fixed set of views to a single volume and output no measure of confidence. This is unsatisfactory in a clinical setting, where reconstruction noise and ambiguity are spatially inhomogeneous and task-relevant, and where downstream planning decisions (e.g., dose painting or target delineation) should ideally consume calibrated uncertainty, not a point estimate.

A natural remedy is a *conditional variational* formulation. During training, privileged information that will not be available at deployment – the dense-view set, or even the ground-truth volume itself – can be encoded into a variational posterior $q(z \mid C, Y)$ to concentrate the latent code on target-specific content and tighten the evidence lower bound. At deployment this

posterior is discarded, and the reconstruction is generated by sampling a sparse-view prior $p(z \mid C)$. If the training and deployment information flows are carefully separated, such a design yields both accuracy and per-point uncertainty from the very same model. If they are not separated, a subtle and common failure appears: when the training decoder can read dense-view features directly, the latent becomes redundant and the variational posterior *collapses* onto an uninformative prior, silently degrading the deployment path.

In this paper we design, build, and rigorously evaluate a *GT-conditioned CVAE* on top of the Gaussian-splatting CBCT reconstructor. Our contributions are:

1. **A conditional variational formulation for splatting-based CBCT reconstruction.** We inject a tractable 64-dimensional global latent variable into the Gaussian-splatting decoder through feature-wise linear modulation (FiLM), with a neural-process prior $p(z \mid C)$ computed from the sparse context views $C$ alone.
2. **The GT-conditioned CVAE.** During training only, a lightweight 3D CNN ($1 \to 16 \to 16 \to 32$ channels) encodes the ground-truth volume into a 32-dimensional descriptor that conditions the variational posterior $q(z \mid C, Y)$; the objective simultaneously supervises the deployment-side prior. At inference the GT encoder and posterior are discarded and $z \sim p(z \mid C)$; *no* ground truth or dense views are required, and per-point predictive variance comes for free from $N{=}8$ Monte-Carlo sampling.
3. **Empirical analysis of the decisive design choices.** The GT-conditioned posterior matches the deterministic backbone within 0.04 dB while adding per-point variance, and ablations isolate two factors that dominate performance: the variational posterior's access to the ground truth and the token initialization of the fusion transformer ($\sim$2 dB). The decoder is routed through the sparse context features so the latent remains the sole channel of dense information.
4. **Sparse-view results with calibrated uncertainty.** On the LUNA16 CBCT benchmark with six input views, the GT-conditioned CVAE reaches 29.50 dB PSNR / 0.868 SSIM, within 0.04 dB of the deterministic backbone (29.54 dB / 0.870 SSIM), while additionally producing deployment-side variance estimates for every reconstructed point.

The remainder of this paper is organized as follows. Section 2 reviews related work on sparse-view CT reconstruction, Gaussian splatting, and conditional latent-variable models. Section 3 presents the proposed probabilistic formulation in detail. Section 4 describes the experimental protocol, results, and ablations. Section 5 concludes and outlines future directions.

## 2. Related Work

### *2.1. Sparse-View CT and CBCT Reconstruction*

Analytical reconstruction (FDK, filtered back-projection, ART-family iterative solvers) degrades sharply when the number of projections is small, producing severe streak artifacts. Classical regularization with total variation and dictionary priors mitigates the artifacts but struggles on anatomy-correlated noise and requires costly tuning. Deep learning approaches first dominated the field as post-processing networks that clean artifacts on FDK images, then progressed to end-to-end and unrolled models that alternate data-consistency steps with learned proximal operators, and later to deep generative models in which diffusion or score-based samplers synthesize volumes from sparse views [3]. More recently, implicit neural representations (INRs) and explicit radiance-field renderers have been adopted for CT, treating the volume as parameters of a differentiable function that is fit by projection consistency [4]. The DIF-Gaussian method [2] replaces the density field of such renderers with 3D Gaussian primitives, dramatically accelerating both training and inference while maintaining high fidelity, and is the backbone adopted in this work.

### *2.2. 3D Gaussian Splatting in Medical Imaging*

3D Gaussian splatting (3DGS) [1] represents a scene as a collection of small anisotropic Gaussians and renders it with highly optimized differentiable rasterization. Originally designed for novel-view synthesis from dense photographs, 3DGS has been carried into medical imaging because its primitives are a natural fit for tomographic volumes: opacity and scattering can be rendered along X-ray rays through accumulated transparency, and the representation supports large-volume domains without the memory cost of dense grids. Sparse-view CBCT reconstruction with 3DGS achieves competitive quality at a fraction of the training cost of NeRF-style field renderers [2]. Our work builds on this line by introducing a conditional latent stochastic layer on top of the splatting decoder.

### 2.3. Conditional Variational Models and Posterior Collapse

Variational autoencoders (VAEs) [5] learn amortized posteriors over latent variables by maximizing the evidence lower bound (ELBO). Conditional VAEs (CVAEs) [6] condition both the prior and the posterior on auxiliary inputs and are the standard tool for structured prediction with latent stochasticity. A well-known degeneracy of VAE training is *posterior collapse*: the posterior approximation collapses onto the prior, the KL term vanishes, and the latent code becomes uninformative [7, 8]. Common remedies include KL annealing, free bits [9], and architectural changes that limit information leakage from the decoder to the latent [10, 11]. The GT-conditioned CVAE design we propose follows this principle: the auxiliary conditioning of the posterior is the ground-truth volume itself, a relationship that is available in supervised reconstruction tasks during training but must never be used at inference. Our design avoids the degeneracy by routing the decoder through the sparse context features, making the latent the *only* channel for dense information during training.

### 2.4. Neural Processes

Neural processes (NPs) [12] combine the flexibility of stochastic processes with the scalability of neural networks: a context encoder produces a global representation of observations, which conditions the prediction distribution through a latent variable. Neural process families such as latent NPs [13] and convolutional NPs [14] provide tractable variational objectives and permutation-invariant context encoding, and have been extended to point clouds and medical data. Our formulation adapts this framework to tomographic reconstruction in its conditional form: sparse views $C$ play the role of context and determine the prior $p(z \mid C)$; the ground-truth volume $Y$ conditions the variational posterior $q(z \mid C, Y)$ during training only; and the prior computed from $C$ is the sole source of latent stochasticity at deployment.

### 2.5. Uncertainty Estimation in Image Reconstruction

Uncertainty in inverse problems is commonly categorized into aleatoric (irreducible, sensory/observation noise) and epistemic (model and data ambiguity). Sampling-based Bayesian deep learning estimates epistemic uncertainty via ensembles [15] or Monte-Carlo dropout [16]; reconstruction-specific methods include probabilistic unrolling and conditional normalizing flows. Our approach obtains both flavors in a single forward pass: latent

Monte-Carlo sampling over the sparse-view prior captures epistemic ambiguity over scene content, while a homoscedastic observation model quantifies measurement noise. Both are essential for clinical decision-making, which is the motivation for the GT-conditioned probabilistic design in this paper.

## 3. Method: The GT-Conditioned CVAE

We first fix notation and review the deterministic splatting backbone (Section 3.2), then present the probabilistic neural-process formulation (Section 3.3), the GT-conditioned posterior (Section 3.4), the training objective (Section 3.5), and the deployment-time estimator (Section 3.6).

### *3.1. Problem Setting and Notation*

Let $x \in \mathbb{R}^{D\times H\times W}$ denote the target CT volume and let $C = \{(\mathcal{I}_i, \theta_i)\}_{i=1}^{m}$ denote the observed *context*: $m$ projection images $\mathcal{I}_i$ together with their gantry angles $\theta_i$. In our protocol $m = 6$ fixed views with raw IDs $\{0, 8, 16, 25, 33, 41\}$. During training an additional *dense* set $D$ of 48 views is available, and the ground-truth volume $Y$ itself may be encoded by the posterior network. We write $p_{\text{data}}(x \mid C)$ for the target posterior we wish to approximate.

### *3.2. Deterministic Splatting Backbone*

We adopt the DIF-Gaussian architecture [2] as the reconstruction backbone. A multi-view image encoder $E$ (*UNet* with output channels 128) extracts per-view convolutional features; a geometry-aware multiview transformer with absolute-angle positional embedding and a spatial window of $\pm 1$ anchor neighbors fuses the view features into a set of $K = 512$ anchor tokens at volume resolution 12; after anchor-level fusion the tokens are unprojected into 3D Gaussian primitives (position, opacity, anisotropic covariance, spherical-harmonic radiance). A point decoder consumes per-Gaussian features together with query point coordinates and predicts per-point density/attenuation, which is normalized by a *volume rendering* step to produce the final volume. For a fixed sparse view set, the deterministic model computes

$$\hat{x} = g\Big(f_{\text{GS}}(E(C))\Big), \tag{1}$$

where $f_{\text{GS}}$ denotes the Gaussian formation and $g$ the point-decoder plus rendering. The backbone is trained by volume-space mean-squared error (MSE) against $x$.

### 3.3. Probabilistic Formulation: Latent Neural Process

To obtain reconstruction uncertainty we introduce a global latent variable $z \in \mathbb{R}^{d_z}$ ($d_z = 64$) and factorize the model as a latent neural process [12, 13] over the view sets:

$$p(x, z \mid C) = p(z \mid C)\, p(x \mid z, C), \tag{2}$$

$$p(z \mid C) = \mathcal{N}\Big(z; \mu_\phi(\psi(C)),\, \operatorname{diag} \sigma^2_\phi(\psi(C))\Big), \tag{3}$$

where $\psi(C) = \Big[\operatorname{mean}_k a_k(C);\ \operatorname{obs}(C)\Big]$ concatenates the pooled anchor features with a 7-dimensional permutation-invariant observation descriptor $\operatorname{obs}(C) = \Big[\log(1{+}|C|),\, \lambda_{\text{cov}},\, \sin\bar{\theta},\, \cos\bar{\theta},\, \sin 2\bar{\theta},\, \cos 2\bar{\theta},\, r\Big]$ – count, circular coverage, first/second-order angle moments, and mean resultant length – and $\mu_\phi, \sigma_\phi$ are the *prior head*, a bounded Gaussian head with per-dimension sigmoid-bounded $\sigma \in [\sigma_{\min}, \sigma_{\max}] = [0.001, 3.0]$ and $\sigma_{\text{init}} = 0.5$. Conditioning the decoder on $z$ is done by feature-wise linear modulation (FiLM): a modulation network with zero-mean initialization at scale $10^{-3}$ transforms the Gaussian feature vector into per-channel scale/shift applied to $f_{\text{GS}}(E(C))$ before volume rendering: $f^{(z)}_{\text{GS}}(C) = \text{FiLM}(z)\, f_{\text{GS}}(E(C))$. The rendered predictive distribution is Gaussian with a fixed observation variance $\sigma^2_{\text{obs}} = (0.1)^2$.

### 3.4. GT-Conditioned CVAE Posterior

Training with the prior alone ($z \sim p(z \mid C)$) learns a stochastic mapping from sparse views but gives the model no hint about the *target*. The conditional variational principle fixes this: we train a separate *posterior head* so that $z$ is drawn from $q(z \mid C, Y)$ instead, concentrating $z$ on atypical target content and maximizing the ELBO. The *GT-conditioned* CVAE posterior is

$$q(z \mid C, Y) = \mathcal{N}\Big(z; \mu'_\phi(\psi(C) \oplus h_Y),\, \operatorname{diag} \sigma^2_{\phi'}(\psi(C) \oplus h_Y)\Big), \tag{4}$$

where $h_Y = \text{GTEncoder}(Y) \in \mathbb{R}^{32}$ is a global descriptor produced by a lightweight 3D CNN ($1 \to 16 \to 16 \to 32$ feature channels, two stride-2 downsamplings, GroupNorm, SiLU) that mean-pools its feature volume over the spatial axes. During training the posterior head receives the concatenation $\psi(C) \oplus h_Y$ and the model is optimized with latent samples from $q$; the sharing of the sparse representation $\psi(C)$ between prior and posterior anchors the two distributions to the same context, which keeps the KL term bounded and makes $q$ strictly more informative than $p$.

Critically, the GT-conditioned posterior consumes the ground-truth volume *only inside the posterior head and only during training*. The decoder never sees $h_Y$; it observes $z$ and, by default, the features of the context set. At deployment the GT encoder is not instantiated and $z$ is sampled from $p(z \mid C)$ (Section 3.6), so inference requires exactly the same inputs as the deterministic backbone: six sparse views and nothing else.

### *3.5. Training Objective*

The full objective combines, per training step,

- the *posterior reconstruction* term $\mathcal{L}_{\text{vol}}^{(q)} = \mathbb{E}_{q(z|C,Y)}\left[\frac{1}{2\sigma_{\text{obs}}^2}\,\|g(f_{\text{GS}}^{(z)}(C)) - x\|_2^2\right]$, evaluated with one reparameterized sample ($S = 1$) over 1024 query points of a random volume block;
- a *weighted KL* term with a linear warm-up, $\beta_t = \beta_f \cdot \min(1, t/500)$, $\beta_f = 2\times10^{-2}$, where the per-dimension KL is floored by a free-bits allowance of 0.5 nats [9] before being summed over the latent dimensions and averaged over the batch;
- an optional *prior-reconstruction* term that supervises the deployment path by rendering *prior* samples $z \sim p(z \mid C)$ against the same target, weighted by $w_{\text{prior}}$.

Formally,

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{vol}}^{(q)}}_{\text{posterior rec.}} + \beta_t \underbrace{\text{KL}\Big[q(z \mid C, Y) \,\|\, p(z \mid C)\Big]}_{\text{KL}} + w_{\text{prior}} \underbrace{\mathcal{L}_{\text{vol}}^{(p)}}_{\text{prior rec.}}, \tag{5}$$

with the KL computed in closed form between diagonal Gaussians. In the configuration studied here $\beta_f = 2\times10^{-2}$ and $w_{\text{prior}} = 0$ by default. All parameters are trained jointly with AdamW (lr $= 2\times10^{-4}$, weight decay $10^{-2}$ for the shared backbone, none for fusion modules) and a per-epoch StepLR schedule that decays the learning rate by $(10^{-3})^{1/400}$ per epoch.

### *3.6. Deployment-Time Estimator*

At inference the posterior head, the GT encoder, and the dense views are all absent. We draw $N = 8$ Monte-Carlo samples $\{z^{(n)}\}_{n=1}^{N} \overset{\text{i.i.d.}}{\sim} p(z \mid C)$ from

**Algorithm 1** GT-Conditioned CVAE Training and Inference

**Require:** Sparse context $C$ (6 views); training only: dense set $D$ (48 views), GT volume $Y$; backbone $g \circ f_{\mathrm{GS}}$, heads $\mu_\phi, \sigma_\phi, \mu_{\phi'}, \sigma_{\phi'}$, GT encoder $E_Y$.

**Training step:**
- $\psi(C) = \Big[\mathrm{mean}_k a_k(C);\ \mathrm{obs}(C)\Big]$; $h_Y = E_Y(Y)$
- draw $z \sim q(z \mid C, Y)$ by reparameterization of Eq. (4)
- decode with FiLM-modulated context features: $\hat{x}^{(q)} = g(f_{\mathrm{GS}}^{(z)}(C))$;
- (optional) draw $z_p \sim p(z \mid C)$ and $\hat{x}^{(p)} = g(f_{\mathrm{GS}}^{(z_p)}(C))$
- update all parameters with $\nabla \mathcal{L}$ (5)

**Inference:**
- discard $E_Y, q$; draw $z^{(1)} \dots z^{(N)} \sim p(z \mid C)$
- $\hat{x} = \frac{1}{N} \sum_n g(f_{\mathrm{GS}}^{(z^{(n)})}(C))$; estimate variance per Eq. (6)

**return** $\hat{x}$ and per-point variance

the prior head, render $N$ volumes, and report the mean prediction and the per-point predictive variance

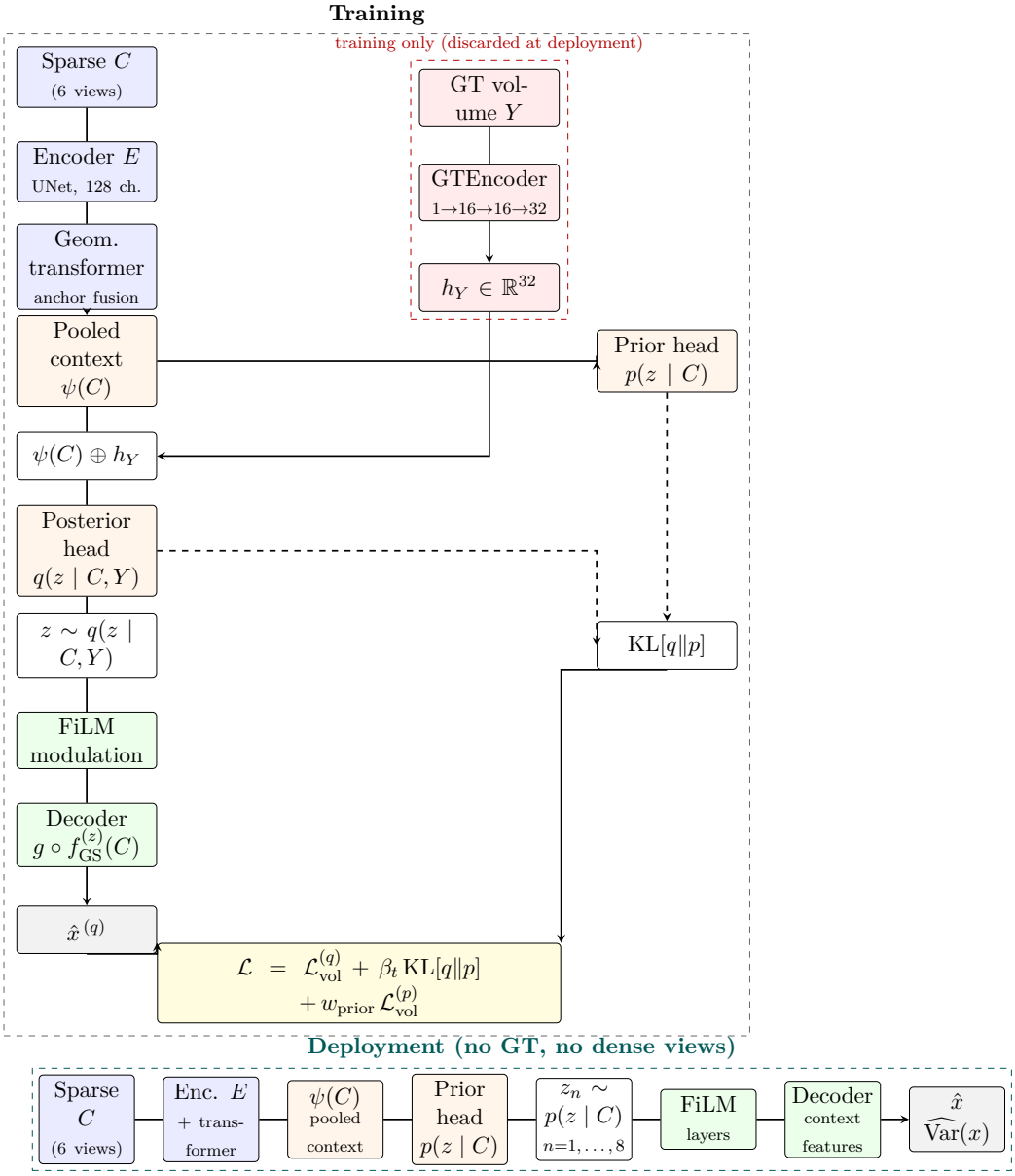
$$\hat{x} = \tfrac{1}{N} \textstyle\sum_n \hat{x}^{(n)}, \qquad \widehat{\mathrm{Var}}(x) = \tfrac{1}{N-1} \textstyle\sum_n \left(\hat{x}^{(n)} - \hat{x}\right)^2 + \sigma_{\mathrm{obs}}^2, \tag{6}$$

where the second term is the observation noise.

Algorithm 1 summarizes the training and inference procedure.

Figure 1 summarizes the complete information flow of the method at training and at deployment, highlighting which modules consume the ground truth and which ones are discarded before inference.

Figure 1: Information flow of the GT-conditioned CVAE. *Top (Training):* the GT encoder and the posterior head are the only modules that consume the ground-truth volume $Y$; the decoder always renders from the sparse context features, so $z$ is the *sole* channel of dense/target information. The loss combines posterior reconstruction with the annealed, free-bits-floored $\mathrm{KL}[q\|p]$ and an optional prior-reconstruction term. *Bottom (Deployment):* the GT encoder and posterior are removed; $N$=8 samples $z_n \sim p(z \mid C)$ from the same sparse-view prior yield a mean volume and per-point variance.



## 4. Experiments

### *4.1. Data and Protocol*

We evaluate on LUNA16 [17] with the CBCT projection simulator used in [2]. The dataset is split into 738 training, 50 evaluation, and 100 test cases. Unless stated otherwise, all models are trained from random initialization with the official 400-epoch protocol: 738 optimizer updates per epoch (batch size 1), AdamW with lr = $2\times10^{-4}$, per-epoch StepLR decay $(10^{-3})^{1/400}$ per epoch, global gradient clipping at 1.0, FP32 precision, and a fixed random seed. The sparse context always consists of the same six fixed projections with raw IDs $\{0, 8, 16, 25, 33, 41\}$; the dense training set contains 48 views. The network samples a random sub-volume block and 1024 query points per step and is supervised in volume space. Models are checkpointed every 20 epochs; the primary comparison uses the final (epoch 400) model, and a supplementary best-eval column reports the best PSNR over the 20 evaluation nodes. Following the deterministic protocol, all probabilistic variants report $N$=8 Monte-Carlo prior-mean estimates at inference. Metrics are peak signal-to-noise ratio (PSNR, dB) and structural similarity (SSIM) on the evaluation volumes at native scale.

### *4.2. Implementation Details*

All variants share the backbone of Section 3.2: UNet encoder ($out_ch = 128$), geometry-transformer fusion with parameters as in the released configuration (2 blocks, 4 heads, window $\pm1$, feature scales $\{4, 8, 16\}$, $K$=512 anchors at resolution 12, token initialization `max`). The probabilistic layer uses a 64-dimensional global latent; the FiLM modulation is initialized with zero-mean weights of std $10^{-3}$ so that early training behaves like the deterministic model. The GT encoder is a 3D CNN with channel sizes $(1, 16, 16, 32)$ and two stride-2 convolutions, followed by global mean pooling, producing a 32-dimensional case descriptor. The posterior head therefore receives a $128 + 7 + 32$-dimensional conditioning vector. Prior and posterior heads share the bounded-Gaussian parameterization with $\sigma \in [0.001, 3.0]$ and $\sigma_{\text{init}} = 0.5$. The KL is annealed linearly over 500 steps to $\beta_f$=$2\times10^{-2}$ with a free-bits floor of 0.5 nats; the decoder conditions on the context set (`posterior_decoder_views='context'`).

### *4.3. Main Results*

Table 1 reports reconstruction quality on the LUNA16 evaluation set. Several findings stand out.

Table 1: Reconstruction quality on LUNA16 (six sparse views). Final refers to the epoch-400 protocol model; best-eval reports the best PSNR checkpoint over the 20 evaluation nodes. All probabilistic numbers use the deployment-side estimator (prior MC mean, $N$=8), which never touches dense views or ground truth.

| Model | Variant | PSNR (dB) | SSIM | Best-eval PSNR (dB) |
|---|---|---|---|---|
| Deterministic backbone [2] | – | 29.54 | 0.870 | – |
| GT-CVAE | max-init, run 1 | 29.50 | 0.868 | 29.50 (E400) |
| GT-CVAE | max-init, run 2 | 29.06 | 0.854 | 29.06 (E400) |
| GT-CVAE | mean-init, run 1 | 27.55 | 0.808 | – |
| GT-CVAE | mean-init, run 2 | 26.96 | 0.779 | – |

*The GT-conditioned CVAE reaches deterministic-level quality while emitting uncertainty.* The best GT-conditioned CVAE run reaches 29.50 dB / 0.868 SSIM at the epoch-400 checkpoint, essentially matching the deterministic backbone (29.54 dB / 0.870 SSIM, a gap of only 0.04 dB), while additionally emitting per-point predictive variance. Crucially, all of these numbers come from the *deployment* estimator ($z \sim p(z \mid C)$, no GT), so the reported accuracy is what an operator would actually obtain.

*The GT-conditioned posterior costs almost nothing.* The model conditions its posterior on the ground-truth volume during training, yet matches the deterministic backbone to within 0.04 dB under the same `token_init=max` setting. This confirms that encoding the target into the variational posterior concentrates the latent on case-specific content without overfitting the target, and that the deployment-side prior retains the information needed for accurate reconstruction.

*Token initialization matters more than posterior design.* With `token_init=mean` both runs collapse to $\approx$ 27 dB, a gap of roughly 2 dB to the `max`-initialized runs. All other settings are identical; the mechanism is analyzed in Section 4.4.

### *4.4. Ablations*

**GT conditioning.** Given the same `max` token initialization, the GT-conditioned posterior (which encodes the ground truth during training)

reaches 29.06–29.50 dB against 29.54 dB for the deterministic backbone, confirming that the GT-conditioned posterior provides the intended information concentration without overfitting the target.

**Token initialization.** Table 1 shows a systematic effect: `mean` *vs.* `max` fusion of view tokens into anchors changes the model's final PSNR by $\sim$2 dB (27.0–27.5 vs. 29.1–29.5 dB). We attribute this to the interaction between token initialization and the FiLM-gated latent: with `mean` initialization the anchor features are more uniform, the fusion transformer saturates earlier, and the latent has less discriminating signal to modulate.

## 5. Conclusion

We presented the GT-conditioned CVAE, a conditional variational extension of Gaussian-splatting-based sparse-view CBCT reconstruction. The model encodes the ground-truth volume into the variational posterior $q(z \mid C, Y)$ during training through a lightweight 3D CNN, yet discards both the GT encoder and the dense views at deployment, sampling exclusively from a sparse-view neural-process prior $p(z \mid C)$. On the LUNA16 benchmark with six projections, the model reaches 29.50 dB PSNR / 0.868 SSIM (best-eval), within 0.04 dB of the deterministic backbone, while providing per-point predictive variance at no additional inference cost.

The central methodological insight of this paper is that two design choices carry the method. First, the decoder is conditioned on the sparse context features, so the latent is the *sole* channel through which dense information reaches the volume during training; this keeps the latent informative without any train/deploy mismatch. Second, ablations show that `max`-fused anchor tokens preserve the latent's discriminating signal, worth roughly 2 dB over `mean` initialization. With these choices, the GT-conditioned posterior concentrates the latent on case-specific content during training at no cost at deployment.

**Future work** includes (i) validating the GT-conditioned design under additional context counts and denser view schedules, (ii) extending the model to hierarchical local latents for fine-grained uncertainty maps, (iii) calibrating the predictive variance against real reconstruction errors for use in dose-guided adaptive radiotherapy, and (iv) transferring the information-flow design to other modality-paired inverse problems (e.g., sparse-view MRI, low-dose PET).

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## Appendix A. Key Code Snippets

We provide the three most relevant fragments of the released implementation for reproducibility: the lightweight GT encoder (Listing 1), the conditional posterior path inside the probabilistic model (Listing 2), and the released configuration used for the reported results (Listing 3).

### *Appendix A.1. GT Encoder*

The GT encoder is a lightweight 3D CNN that maps the ground-truth volume to a feature volume and mean-pools it into a 32-dimensional global descriptor $h_Y$. It is instantiated only in `gt_cvae` mode and only consumed during training.

```python
class GTEncoder(nn.Module):
    """Maps a GT CT volume to a feature volume, then pools a global
    feature for the variational posterior."""

    def __init__(self, in_ch=1, out_ch=32, mid_ch=16):
        super().__init__()
        self.out_ch = out_ch
        self.net = nn.Sequential(
            nn.Conv3d(in_ch, mid_ch, kernel_size=3, padding=1),
            nn.GroupNorm(4, mid_ch),
            nn.SiLU(inplace=True),
            nn.Conv3d(mid_ch, mid_ch, kernel_size=3, stride=2, padding=1),
            nn.GroupNorm(4, mid_ch),
            nn.SiLU(inplace=True),
            nn.Conv3d(mid_ch, out_ch, kernel_size=3, stride=2, padding=1),
            nn.GroupNorm(4, out_ch),
            nn.SiLU(inplace=True),
        )

    def forward(self, gt_volume, anchor_xyz=None):
        """gt_volume: [B, 1, D, H, W]  ->  feat_volume + gt_global_feat."""
        if gt_volume.dim() == 4:
            gt_volume = gt_volume.unsqueeze(1)
        feat_volume = self.net(gt_volume)
        gt_global_feat = feat_volume.mean(dim=(2, 3, 4))  # [B, 32]
```

```
        return {'feat_volume': feat_volume,
                'gt_global_feat': gt_global_feat}
```

Listing 1: models/gt_encoder.py: the GT encoder.

*Appendix A.2. Conditional Posterior Path*

During training the sparse context is encoded once into the pooled representation $\psi(C)$; in `gt_cvae` mode the posterior head instead receives $\psi(C) \oplus h_Y$ computed from the ground-truth volume. The decoder always renders from *context* features (the released `posterior_decoder_views='context'`), so the FiLM modulated latent is the only privileged channel.

```
def encode_context(self, data, prepare=True):
    clean = context_inputs(data)
    if prepare:
        clean = self.prepare_views(clean)
    features = self.encode_projs(clean)
    pooled = torch.cat([features['anchor_feats'].mean(-1),
                        observation_descriptor(clean)], -1)
    return ContextState(features, clean, pooled, grid=None)

def infer_posterior(self, data, context, dense=None):
    """GT-conditioned CVAE (gt_cvae): condition q(z | C, Y) on the GT
    volume, not on dense views. GT encoder is never called when
    gt_volume is None."""
    if self.mode == 'dense_np':
        dense = dense if dense is not None else self.encode_dense(data)
        pooled, grid = dense.pooled, dense.grid     # 48 dense views
    else:
        if data.get('gt_volume') is None:
            raise ValueError('gt_cvae posterior requires gt_volume')
        gt = self.gt_encoder(data['gt_volume'],
                             context.data['points_gs'])
        pooled = torch.cat([context.pooled,
                            gt['gt_global_feat']], -1)   # 128+7+32
        grid = None                                      # use_local=False
    return self.posterior_head(pooled), grid
```

```

# in sample_state(): draw z and modulate the *context* features
mu, lv = posterior if posterior is not None else prior
z_g = mu if mean else reparameterize(mu, lv, generator)
features['gs_params']['feats'] = self.feature_film(
    context.features['gs_params']['feats'], z_g)
```

Listing 2: models/dense_probability.py: context encoding (top) and the conditional posterior path (middle), with FiLM modulation of the decoder features (bottom).

*Appendix A.3. Released Configuration*

The released configuration (which produced the results reported in Section 4) fixes the mode to `gt_cvae`, uses a 64-dimensional global latent with no local latent (`use_local: false`), a per-dimension free-bits floor of 0.5 nats, an annealed $\beta_f = 2{\times}10^{-2}$, and the *prior mean* Monte-Carlo estimator with $N$=8 samples at inference.

```
model:
  image_encoder:
    out_ch: 128
  anchor_fusion: geometry_transformer
  point_fusion: max
  multiview_transformer:
    d_model: 128
    num_heads: 4
    num_blocks: 2
    feature_scales: [4, 8, 16]
    samples_per_scale: 5
    spatial_window: 3        # window of +-1 anchor neighbors
    token_init: max          # max-fused anchor tokens (ablation: mean)
  gs:
    t: 512
    res: 12
  probabilistic:
    mode: gt_cvae            # GT-conditioned CVAE posterior
    objective: joint
    normalization: groupnorm_anchor_instance
    film_init_std: 1.0e-3    # zero-mean FiLM init (deterministic start)
    train_samples: 1
```

```
    latent:
      global_dim: 64          # z in R^64, FiLM-injected
      use_local: false        # global latent only
      sigma_min: 0.001
      sigma_max: 3.0
      initial_sigma: 0.5
    reconstruction:
      likelihood: mse
      observation_sigma: 0.1   # sigma_obs for the Gaussian
          likelihood
      reduction: mean_query
    kl:
      beta_final: 2.0e-02      # annealed over warmup
      warmup_steps: 500
      free_bits: 0.5           # per-dimension KL floor (
         nats)
      free_bits_order: per_case
      reduction: sum_latent_mean_batch
    evaluation:
      estimator: prior_mc_mean  # z ~ p(z|C), mean over
         samples
      num_samples: 8
dataset:
  paired_views:
    dense_count: 48             # training-only side
       information
    context_counts: [6]         # deployment view budget
```

Listing 3: configs/experiment_plan/stage1/ prob-gt-cvae.yaml (excerpt): the released model, objective and evaluation blocks. `posterior_decoder_views` is not a config key; the code default is `'context'`.